

	MPS Engineering Work Order for In-kind Replacement (Non MOC) Инженерная Инструкция для Работ MPS по Аналогичной Замене (Без КЗИ)											
	Anomaly Report: Отчет об аномалии:	N/A	EWR #: # ЗИР:	25-0112	Subsystem: Подсистема:	SL	Line №: № Линии:	63-9100-SL-5038-3/4"-150H22-HC				
	Area: Участок:	SGP	WO: РЗ:	1704523	P&ID #: СТИКИП	62-0700-B-PID-0741-10	Title: Название:	TA26 TLR#23141 REMOVAL ON TL LINE				
TENGIZCHEVROIL ТЕНГИЗШЕВРОЙЛ	Unit: Установка:	U700	Drawing/Sketch No: Чертеж/Схема No:		L-SK-001		TCO Approved WPS: Утвержденная Сварочная ПроцедураТШО:		20AW			
Standard of Acceptance: Стандарт Приемки:	B31.3		TCO Specification: Спецификации ТШО:		W-ST-2025		Is wall thickness measurement required at tie-in points prior to start works (to verify existing pipe sound wall thickness) Требуется ли замер толщины стенки на точках врезки перед началом работ (для определения приемлемой толщины существующего трубопровода)		YES			
						Is 100% PT/MT required for weld bevel ends prior to welding Требуется ли 100% ККД/МПД для проверки кромок перед сваркой		YES/ДА				
Shop Hydro-test Испытание Гидро-тестом в Цеху		NA/НП barg		Piping NDT НК Трубопровода		NA/НП VT/BK 100%		Coating System #: № Системы Покрытия:		12.1	Hydrogen Bake out Водородный Отжиг	NA/НП
Field Hydro-test Испытание Гидро-тестом на Участке		NA/НП barg		Piping Support Structural NDT НК Опорной Конструкции Трубопровода		NA/НП		Insulation type and thickness: Тип и толщина изоляции:		HC 40mm	PMI for Alloy materials (SP 20) ХАМ для Материалов из сплавов (ПТБ 20)	NA/НП
HARDNESS TESTING (Maximum up to 200HB) ОПРЕДЕЛЕНИЕ ТВЕРДОСТИ (Максимум до 200HB)		NA/НП		Stress Relief (PWHT) Снятие Напряжений (ПСТО)		NA/НП		Notes Примечания		SERVICE TEST		

Job Description
Описание работ

Please, see next pages for step associated with work execution/ Прошу ссылаться на следующие страницы по выполнению работ.

Approved by Потверждено		Name Имя	Signature Подпись
Originated by: Подготовлено:	DE MPS Engineer Инженер DE MPS	Adil Maxutov	Adil Maxutov Digitally signed by Adil Maxutov Date: 2025.03.02 09:13:15 +05'00'
Approved by: Утверждено:	DE Engineer-Consultant DE Инженер-Консультант проектной группы		Bekzat Abzaliyev Digitally signed by Bekzat Abzaliyev Date: 2025.03.13 13:42:31 +05'00'
Reviewed by: Рассмотрено:	RE QM Lead: Вед. Специалист КК Отдела Надежности:		
Reviewed by: Рассмотрено:	FER Authorized Inspector: Уполномоченный инспектор НСО:		
Reviewed by: Рассмотрено:	TA Group Supervisor: Руководитель группы КР:		
Reviewed by: Рассмотрено:	TA Unit Coordinator: Координатор КР установок ОЗ:		
Approved by: Утверждено:	Design Engineering Supervisor: Руководитель инженерно-проектной группы:		

PRESENT SITUATION:

- There is a steam leak on pipe section of SL steam line 63-9100-SL-5038-3/4"-150H22-HC on U700. TLR# 23141, WO# 1704523.

PROPOSED SOLUTION:

- It is required to replace damaged pipe section of SL steam line 63-9100-SL-5038-3/4"-150H22-HC.

PRELIMINARY SITE WORK:

- Prior to the start of the mechanical activities, perform wall thickness testing at Tie-in Points (TP) showed in sketch drawing L-SK-001 to confirm that wall thickness is adequate for welding.

NOTE: Results shall be reviewed and approved by the Fixed Equipment Reliability (FER) Unit Inspector.

- Prior to commencing any fabrication works verify dimensions on site.

SHOP WORKS:

- Withdraw materials required for this EWO from TCO Warehouse according to Material Request(s).
- Abrasive blast and coat new piping, as specified on sketch drawing L-SK-001 in accordance with TES COM-SU-5191-TCO.

SITE WORKS:

- Perform relevant job safety assessments (PPHA & JSA) and obtain required Permits-to-Work (PTWs).
- Demarcate work area as necessary.

NOTE: TCO Operations shall isolate, depressurize, drain and ready for Hot Work, Equipment and Piping associated with this Project, if and as applicable, and shall hand it over to CONTRACTOR; fully accessible, safe and Locked-Out-Tagged-Out (LOTO) as scheduled and agreed between TCO Operations and CONTRACTOR.

- Remove all cladding and insulation to the extent required.
- Cut and remove existing piping shown on "Destruct Detail" of sketch drawing L-SK-001.
- Bevel cut ends of existing piping at the tie-in points for field welding as shown on sketch drawing L-SK-001.
- Inspect the prepared weld bevel ends 100% VT and 100 % MT/PT for defects.

NOTE: Perform inspection of weld bevels PRIOR to welding.

- Install piping as specified on sketch drawing L-SK-001. For field welds use TCO-approved WPS indicated in on sketch drawing.

- Perform NDE of field welds as specified in on sketch drawing L-SK-001.
- Perform surface preparation and touch up field welds and any other coating damages in accordance with TES COM-SU-4743-TCO and coating system(s) per TES COM-SU-5191-TCO specified on drawings.
- All QM piping passport documentation (A-category Checklists) to be verified and signed by TCO-approved QM Inspector, but not later than 24 hours after mechanical scope completion.
- Advise Mursal Rauan / Ramazanov Nurbulat DE TA26 SGP/SGI Mechanical Engineers at ext. 4259 about work completion and invite all necessary people to PSSR.
- Job Pack Coordinator shall submit the completed PIC checklist to Operations department after being signed by Quality Management group. (Form is provided by QM Dept.).
- Conduct PSSR and close out A-category punch points.
- Reinstate insulation and cladding as per IRM-SU-1381 as specified in EWO.
- Close out all B-category punch points
- Clean up the work area and demobilize from site.

AFTER COMPLETION OF WORK:

- Perform service test.
- Final as-built package (including B-category Checklist) shall be submitted to TCO QM group within 2 weeks after completion of the whole scope of works.

REFERENCED TCO SPECS:

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| • A-ST-2008 | Basic Engineering Data |
| • SI-105 | Permit to work and hazard analysis |
| • W-ST-2025 | Welding, PWHT and NDE of piping |
| • PIM-SU-2505-TCO | Carbon Steel Piping Fabrication |
| • RM-SU-1381-TCO | Thermal Insulation for Hot Lines |